

Nilesh Rawal

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SUMMARY

Embedded Software Engineer with hands-on experience in platform bring-up, device driver development, and system integration across Embedded Linux, QNX, and Android on ARM64 and x86 platforms. Skilled in kernel-level debugging, network driver development (XGMAC on QNX), and Android display bring-up on virtual ARM SoCs. Experienced in diagnosing and resolving issues across the kernel-to-userspace stack, optimizing networking and system performance.

EXPERIENCE

Embedded Software Engineer

Vayavya Labs Pvt. Ltd.

Jul 2022 – Present

Belagavi, IN

- Developed and integrated device drivers (Ethernet, UART, SPI) across QNX and Linux platforms on ARM64 and x86 architectures.
- Contributed to Android platform bring-up on virtual ARM64 SoC, focusing on kernel-to-userspace debugging and issue resolution across JNI, DRM, and graphics layers.
- Worked on EDMA (10G NIC) driver development across QNX and Linux, covering initialization, data path implementation, and performance optimization.
- Build and customize Buildroot and Android build environments for platform adaptation.
- Performed driver testing, debugging, and performance analysis using tools like GDB, strace, ftrace, and ktrace, ensuring stability and throughput optimization.

KEY PROJECTS

- **Android Display & Platform Bring-up – ARM Cortex Virtual SoC:** Brought up Android display subsystem from kernel to HAL and userspace integration. Debugged kernel-level graphics pipeline and stabilized frame rendering and buffer management using **libdrm** and **OpenGL ES**. Integrated display stack with DRM, HAL, and Android framework layers. Customized Buildroot and AOSP builds to reproduce and debug boot and graphics failures. Performed low-level debugging (assembly, SoC timing, and configuration) to resolve hardware-level issues. Partnered with firmware and verification teams for validation and regression testing.
- **EDMA / 10GbE Driver Development – QNX 7.1 (ARM64):** Designed and implemented an EDMA-accelerated 10GbE Ethernet driver using **io-sock** and **iflib**. Ported core network stack mechanisms (interrupt handling via MSI-X, DMA descriptor management) from Linux to QNX. Implemented optimized buffer allocation and scatter-gather DMA for high throughput. Refactored driver for direct network stack integration, reducing software overhead. Validated DMA and interrupt performance using profiling and system-level testing on ARM64 hardware. Developed automated test suites to verify TX/RX data path efficiency and reliability.

TECHNICAL SKILLS

Languages: C, C++, Embedded C, Shell Script

Platforms: QNX, FreeBSD, Embedded Linux, Android (AOSP)

Drivers: UART, Ethernet (io-sock, iflib), Display (libdrm, HAL)

Build Systems: GCC, QCC, Buildroot, Yocto, U-Boot, AOSP

Tools: Git, GDB, KGDB, strace, ftrace, perf, objdump

Debugging: LLDB, Perfetto, ktrace, JTAG, Segger J-Link

Protocols: UART, SPI, I2C, CAN, VLAN, ARP, TCP/IP

EDUCATION

Post Graduate Diploma in Embedded Systems and Design

2023

Centre for Development of Advanced Computing (C-DAC)

Pune, IN

Bachelor of Technology in Electrical Engineering

2022

Sant Gajanan Maharaj College of Engineering

Mahagaon, IN